

Newman-Penrose Formalism Calculations

Stephani et al., Equation (18.50)

Metric

$$ds^2 = \left(-\frac{\lambda r^p}{B^2 + r^{2p}} \right) dt \otimes dt + \left(\frac{B\lambda p r^p x}{B^2 + r^{2p}} \right) dt \otimes dy + \left(\frac{B^2 + r^{2p}}{\lambda r^p} \right) dr \otimes dr + \left(\frac{1}{2} (B^2 r + r^{2p+1}) r^{-p-\sqrt{p+1}\sqrt{-p+1}} \right) dx \otimes dx + \left(\frac{B\lambda p r^p x}{B^2 + r^{2p}} \right) dy \otimes dt + \left(-\frac{2 B^2 \lambda p^2 r^{2p} x^2 - (B^4 r + 2 B^2 r^{2p+1} + r^{4p+1}) r^{\sqrt{p+1}\sqrt{-p+1}}}{2 (B^2 r^p + r^{3p})} \right) dy \otimes dy$$

Coordinates

$$(x^\mu) = (t, r, x, y)$$

Null Tetrad

Covariant components

$$l = \left(\sqrt{\frac{1}{2}} \sqrt{\lambda} \sqrt{\frac{r^p}{B^2 + r^{2p}}} \right) dt + \left(-\frac{1}{2} \sqrt{2} B \sqrt{\lambda} p x \sqrt{\frac{r^p}{B^2 + r^{2p}}} + \frac{1}{2} \sqrt{2} \sqrt{\frac{1}{2}} \sqrt{\frac{B^2 r r^{\sqrt{-p^2+1}}}{r^p} + \frac{r r^{2p} r^{\sqrt{-p^2+1}}}{r^p}} \right) dy$$

$$n = \left(\sqrt{\frac{1}{2}} \sqrt{\lambda} \sqrt{\frac{r^p}{B^2 + r^{2p}}} \right) dt + \left(-\frac{1}{2} \sqrt{2} B \sqrt{\lambda} p x \sqrt{\frac{r^p}{B^2 + r^{2p}}} - \frac{1}{2} \sqrt{2} \sqrt{\frac{1}{2}} \sqrt{\frac{B^2 r r^{\sqrt{-p^2+1}}}{r^p} + \frac{r r^{2p} r^{\sqrt{-p^2+1}}}{r^p}} \right) dy$$

$$m = \left(\frac{\sqrt{\frac{1}{2}} \sqrt{\frac{B^2}{r^p} + \frac{r^{2p}}{r^p}}}{\sqrt{\lambda}} \right) dr + \left(\frac{i \sqrt{\frac{B^2}{r^p} + \frac{r^{2p}}{r^p}} \sqrt{r}}{2 r^{\frac{1}{2}} \sqrt{-p^2+1}} \right) dx$$

$$\bar{m} = \left(\frac{\sqrt{\frac{1}{2}} \sqrt{\frac{B^2}{r^p} + \frac{r^{2p}}{r^p}}}{\sqrt{\lambda}} \right) dr + \left(-\frac{i \sqrt{\frac{B^2}{r^p} + \frac{r^{2p}}{r^p}} \sqrt{r}}{2 r^{\frac{1}{2}} \sqrt{-p^2+1}} \right) dx$$

Contravariant components

$$l = \left(\frac{Bpr^p r^{\frac{1}{2}} \sqrt{p+1} \sqrt{-p+1} x}{\sqrt{B^2 + r^{2p}} \sqrt{r} r^{\frac{1}{2}p} r^{\frac{1}{2}p} \sqrt{p+1} \sqrt{-p+1}} - \frac{\sqrt{2} B^2}{2 \sqrt{B^2 + r^{2p}} \sqrt{\lambda} r^{\frac{1}{2}p}} - \frac{\sqrt{2} r^{2p}}{2 \sqrt{B^2 + r^{2p}} \sqrt{\lambda} r^{\frac{1}{2}p}} \right) dt + \left(\frac{r^{\frac{1}{2}p}}{\sqrt{B^2 + r^{2p}} \sqrt{r} r^{\frac{1}{2}} \sqrt{p+1} \sqrt{-p+1}} \right) dy$$

$$n = \left(-\frac{Bpr^p r^{\frac{1}{2}} \sqrt{p+1} \sqrt{-p+1} x}{\sqrt{B^2 + r^{2p}} \sqrt{r} r^{\frac{1}{2}p} r^{\frac{1}{2}p} \sqrt{p+1} \sqrt{-p+1}} - \frac{\sqrt{2} B^2}{2 \sqrt{B^2 + r^{2p}} \sqrt{\lambda} r^{\frac{1}{2}p}} - \frac{\sqrt{2} r^{2p}}{2 \sqrt{B^2 + r^{2p}} \sqrt{\lambda} r^{\frac{1}{2}p}} \right) dt + \left(-\frac{r^{\frac{1}{2}p}}{\sqrt{B^2 + r^{2p}} \sqrt{r} r^{\frac{1}{2}} \sqrt{p+1} \sqrt{-p+1}} \right) dy$$

$$m = \left(\frac{\sqrt{B^2 + r^{2p}} \sqrt{\lambda} r^{\frac{1}{2}p}}{\sqrt{2} B^2 + \sqrt{2} r^{2p}} \right) dr + \left(\frac{i r^{\frac{1}{2}p} r^{\frac{1}{2}p} \sqrt{p+1} \sqrt{-p+1}}{\sqrt{B^2 + r^{2p}} \sqrt{r}} \right) dx$$

$$\bar{m} = \left(\frac{\sqrt{B^2 + r^{2p}} \sqrt{\lambda} r^{\frac{1}{2}p}}{\sqrt{2} B^2 + \sqrt{2} r^{2p}} \right) dr + \left(-\frac{i r^{\frac{1}{2}p} r^{\frac{1}{2}p} \sqrt{p+1} \sqrt{-p+1}}{\sqrt{B^2 + r^{2p}} \sqrt{r}} \right) dx$$

Directional Derivatives

$$D = \frac{Bpr^p r^{\frac{1}{2}} \sqrt{p+1} \sqrt{-p+1} x}{\sqrt{B^2 + r^{2p}} \sqrt{r} r^{\frac{1}{2}p} r^{\frac{1}{2}p} \sqrt{p+1} \sqrt{-p+1}} - \frac{\sqrt{2} B^2}{2 \sqrt{B^2 + r^{2p}} \sqrt{\lambda} r^{\frac{1}{2}p}} - \frac{\sqrt{2} r^{2p}}{2 \sqrt{B^2 + r^{2p}} \sqrt{\lambda} r^{\frac{1}{2}p}} \partial_t + \frac{r^{\frac{1}{2}p}}{\sqrt{B^2 + r^{2p}} \sqrt{r} r^{\frac{1}{2}} \sqrt{p+1} \sqrt{-p+1}} \partial_y$$

$$\Delta = -\frac{Bpr^p r^{\frac{1}{2}} \sqrt{p+1} \sqrt{-p+1} x}{\sqrt{B^2 + r^{2p}} \sqrt{r} r^{\frac{1}{2}p} r^{\frac{1}{2}p} \sqrt{p+1} \sqrt{-p+1}} - \frac{\sqrt{2} B^2}{2 \sqrt{B^2 + r^{2p}} \sqrt{\lambda} r^{\frac{1}{2}p}} - \frac{\sqrt{2} r^{2p}}{2 \sqrt{B^2 + r^{2p}} \sqrt{\lambda} r^{\frac{1}{2}p}} \partial_t - \frac{r^{\frac{1}{2}p}}{\sqrt{B^2 + r^{2p}} \sqrt{r} r^{\frac{1}{2}} \sqrt{p+1} \sqrt{-p+1}} \partial_y$$

$$\delta = \frac{\sqrt{B^2 + r^{2p}} \sqrt{\lambda} r^{\frac{1}{2}p}}{\sqrt{2} B^2 + \sqrt{2} r^{2p}} \partial_r + \frac{i r^{\frac{1}{2}p} r^{\frac{1}{2}p} \sqrt{p+1} \sqrt{-p+1}}{\sqrt{B^2 + r^{2p}} \sqrt{r}} \partial_x$$

$$\bar{\delta} = \frac{\sqrt{B^2 + r^{2p}} \sqrt{\lambda} r^{\frac{1}{2}p}}{\sqrt{2} B^2 + \sqrt{2} r^{2p}} \partial_r - \frac{i r^{\frac{1}{2}p} r^{\frac{1}{2}p} \sqrt{p+1} \sqrt{-p+1}}{\sqrt{B^2 + r^{2p}} \sqrt{r}} \partial_x$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\begin{aligned} \kappa = & -\frac{\sqrt{B^2 + r^{2p}} B^2 \sqrt{\lambda} p r^{\frac{1}{2}p}}{2(\sqrt{2}B^4 r + 2\sqrt{2}B^2 r r^{2p} + \sqrt{2}r r^4 p)} + \frac{\sqrt{B^2 + r^{2p}} B^2 \sqrt{\lambda} \sqrt{p+1} \sqrt{-p+1} r^{\frac{1}{2}p}}{4(\sqrt{2}B^4 r + 2\sqrt{2}B^2 r r^{2p} + \sqrt{2}r r^4 p)} - \frac{i \sqrt{B^2 + r^{2p}} B \sqrt{\lambda} p r^{\frac{3}{2}p}}{\sqrt{2}B^4 r + 2\sqrt{2}B^2 r r^{2p} + \sqrt{2}r r^4 p} + \frac{\sqrt{B^2 + r^{2p}} B^2 \sqrt{\lambda} r^{\frac{1}{2}p}}{4(\sqrt{2}B^4 r + 2\sqrt{2}B^2 r r^{2p} + \sqrt{2}r r^4 p)} \\ & + \frac{\sqrt{B^2 + r^{2p}} \sqrt{\lambda} p r^{\frac{5}{2}p}}{2(\sqrt{2}B^4 r + 2\sqrt{2}B^2 r r^{2p} + \sqrt{2}r r^4 p)} + \frac{\sqrt{B^2 + r^{2p}} \sqrt{\lambda} \sqrt{p+1} \sqrt{-p+1} r^{\frac{5}{2}p}}{4(\sqrt{2}B^4 r + 2\sqrt{2}B^2 r r^{2p} + \sqrt{2}r r^4 p)} + \frac{\sqrt{B^2 + r^{2p}} \sqrt{\lambda} r^{\frac{5}{2}p}}{4(\sqrt{2}B^4 r + 2\sqrt{2}B^2 r r^{2p} + \sqrt{2}r r^4 p)} \end{aligned}$$

$$\mu = 0$$

$$\lambda = 0$$

$$\begin{aligned} \nu = & \frac{\sqrt{B^2 + r^{2p}} B^2 \sqrt{\lambda} p r^{\frac{1}{2}p}}{2(\sqrt{2}B^4 r + 2\sqrt{2}B^2 r r^{2p} + \sqrt{2}r r^4 p)} - \frac{\sqrt{B^2 + r^{2p}} B^2 \sqrt{\lambda} \sqrt{p+1} \sqrt{-p+1} r^{\frac{1}{2}p}}{4(\sqrt{2}B^4 r + 2\sqrt{2}B^2 r r^{2p} + \sqrt{2}r r^4 p)} + \frac{i \sqrt{B^2 + r^{2p}} B \sqrt{\lambda} p r^{\frac{3}{2}p}}{\sqrt{2}B^4 r + 2\sqrt{2}B^2 r r^{2p} + \sqrt{2}r r^4 p} - \frac{\sqrt{B^2 + r^{2p}} B^2 \sqrt{\lambda} r^{\frac{1}{2}p}}{4(\sqrt{2}B^4 r + 2\sqrt{2}B^2 r r^{2p} + \sqrt{2}r r^4 p)} \\ & - \frac{\sqrt{B^2 + r^{2p}} \sqrt{\lambda} p r^{\frac{5}{2}p}}{2(\sqrt{2}B^4 r + 2\sqrt{2}B^2 r r^{2p} + \sqrt{2}r r^4 p)} - \frac{\sqrt{B^2 + r^{2p}} \sqrt{\lambda} \sqrt{p+1} \sqrt{-p+1} r^{\frac{5}{2}p}}{4(\sqrt{2}B^4 r + 2\sqrt{2}B^2 r r^{2p} + \sqrt{2}r r^4 p)} - \frac{\sqrt{B^2 + r^{2p}} \sqrt{\lambda} r^{\frac{5}{2}p}}{4(\sqrt{2}B^4 r + 2\sqrt{2}B^2 r r^{2p} + \sqrt{2}r r^4 p)} \end{aligned}$$

$$\begin{aligned} \alpha = & \frac{\sqrt{2} \sqrt{B^2 + r^{2p}} B^2 \sqrt{\lambda} p r^{\frac{1}{2}p}}{8(B^4 r + 2B^2 r r^{2p} + r r^4 p)} + \frac{\sqrt{2} \sqrt{B^2 + r^{2p}} B^2 \sqrt{\lambda} \sqrt{p+1} \sqrt{-p+1} r^{\frac{1}{2}p}}{8(B^4 r + 2B^2 r r^{2p} + r r^4 p)} + \frac{i \sqrt{2} \sqrt{B^2 + r^{2p}} B \sqrt{\lambda} p r^{\frac{3}{2}p}}{4(B^4 r + 2B^2 r r^{2p} + r r^4 p)} \\ & - \frac{\sqrt{2} \sqrt{B^2 + r^{2p}} B^2 \sqrt{\lambda} r^{\frac{1}{2}p}}{8(B^4 r + 2B^2 r r^{2p} + r r^4 p)} - \frac{\sqrt{2} \sqrt{B^2 + r^{2p}} \sqrt{\lambda} p r^{\frac{5}{2}p}}{8(B^4 r + 2B^2 r r^{2p} + r r^4 p)} + \frac{\sqrt{2} \sqrt{B^2 + r^{2p}} \sqrt{\lambda} \sqrt{p+1} \sqrt{-p+1} r^{\frac{5}{2}p}}{8(B^4 r + 2B^2 r r^{2p} + r r^4 p)} - \frac{\sqrt{2} \sqrt{B^2 + r^{2p}} \sqrt{\lambda} r^{\frac{5}{2}p}}{8(B^4 r + 2B^2 r r^{2p} + r r^4 p)} \end{aligned}$$

$$\begin{aligned} \beta = & -\frac{\sqrt{2} \sqrt{B^2 + r^{2p}} B^2 \sqrt{\lambda} p r^{\frac{1}{2}p}}{8(B^4 r + 2B^2 r r^{2p} + r r^4 p)} - \frac{\sqrt{2} \sqrt{B^2 + r^{2p}} B^2 \sqrt{\lambda} \sqrt{p+1} \sqrt{-p+1} r^{\frac{1}{2}p}}{8(B^4 r + 2B^2 r r^{2p} + r r^4 p)} - \frac{i \sqrt{2} \sqrt{B^2 + r^{2p}} B \sqrt{\lambda} p r^{\frac{3}{2}p}}{4(B^4 r + 2B^2 r r^{2p} + r r^4 p)} \\ & + \frac{\sqrt{2} \sqrt{B^2 + r^{2p}} B^2 \sqrt{\lambda} r^{\frac{1}{2}p}}{8(B^4 r + 2B^2 r r^{2p} + r r^4 p)} + \frac{\sqrt{2} \sqrt{B^2 + r^{2p}} \sqrt{\lambda} p r^{\frac{5}{2}p}}{8(B^4 r + 2B^2 r r^{2p} + r r^4 p)} - \frac{\sqrt{2} \sqrt{B^2 + r^{2p}} \sqrt{\lambda} \sqrt{p+1} \sqrt{-p+1} r^{\frac{5}{2}p}}{8(B^4 r + 2B^2 r r^{2p} + r r^4 p)} + \frac{\sqrt{2} \sqrt{B^2 + r^{2p}} \sqrt{\lambda} r^{\frac{5}{2}p}}{8(B^4 r + 2B^2 r r^{2p} + r r^4 p)} \end{aligned}$$

$$\tau = -\frac{\sqrt{B^2 + r^{2p}} \sqrt{\lambda} \sqrt{p+1} \sqrt{-p+1} r^{\frac{1}{2}p}}{4(\sqrt{2}B^2 r + \sqrt{2}r r^{2p})} - \frac{\sqrt{B^2 + r^{2p}} \sqrt{\lambda} r^{\frac{1}{2}p}}{4(\sqrt{2}B^2 r + \sqrt{2}r r^{2p})}$$

$$\pi = \frac{\sqrt{B^2 + r^{2\,p}}\sqrt{\lambda}\sqrt{p+1}\sqrt{-p+1}r^{\frac{1}{2}\,p}}{4\left(\sqrt{2}B^2r + \sqrt{2}rr^{2\,p}\right)} + \frac{\sqrt{B^2 + r^{2\,p}}\sqrt{\lambda}r^{\frac{1}{2}\,p}}{4\left(\sqrt{2}B^2r + \sqrt{2}rr^{2\,p}\right)}$$

$$\epsilon = 0$$

$$\gamma = 0$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = 0$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = 0$$

$$\Lambda = 0$$

Weyl Tensor Components

$$\begin{aligned}
\Psi_0 = & \frac{3 B^4 \lambda p^2 r^p}{8 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)} + \frac{B^4 \lambda \sqrt{p+1} p \sqrt{-p+1} r^p}{8 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)} + \frac{3 i B^3 \lambda p^2 r^2 p}{2 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)} \\
& + \frac{i B^3 \lambda \sqrt{p+1} p \sqrt{-p+1} r^2 p}{4 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)} - \frac{3 B^4 \lambda p r^p}{8 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)} - \frac{9 B^2 \lambda p^2 r^3 p}{4 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)} \\
& - \frac{3 i B^3 \lambda p r^2 p}{4 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)} - \frac{3 i B \lambda p^2 r^4 p}{2 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)} \\
& + \frac{i B \lambda \sqrt{p+1} p \sqrt{-p+1} r^4 p}{4 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)} + \frac{3 \lambda p^2 r^5 p}{8 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)} \\
& - \frac{\lambda \sqrt{p+1} p \sqrt{-p+1} r^5 p}{8 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)} - \frac{3 i B \lambda p r^4 p}{4 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)} + \frac{3 \lambda p r^5 p}{8 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)}
\end{aligned}$$

$$\Psi_1 = 0$$

$$\begin{aligned}
\Psi_2 = & -\frac{B^4 \lambda p^2 r^p}{8 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)} + \frac{B^4 \lambda \sqrt{p+1} p \sqrt{-p+1} r^p}{8 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)} - \frac{i B^3 \lambda p^2 r^2 p}{2 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)} \\
& + \frac{i B^3 \lambda \sqrt{p+1} p \sqrt{-p+1} r^2 p}{4 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)} + \frac{B^4 \lambda p r^p}{8 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)} \\
& + \frac{3 B^2 \lambda p^2 r^3 p}{4 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)} + \frac{i B^3 \lambda p r^2 p}{4 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)} + \frac{i B \lambda p^2 r^4 p}{2 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)} \\
& + \frac{i B \lambda \sqrt{p+1} p \sqrt{-p+1} r^4 p}{4 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)} - \frac{\lambda p^2 r^5 p}{8 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)} - \frac{\lambda \sqrt{p+1} p \sqrt{-p+1} r^5 p}{8 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)} \\
& + \frac{i B \lambda p r^4 p}{4 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)} - \frac{\lambda p r^5 p}{8 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)}
\end{aligned}$$

$$\Psi_3 = 0$$

$$\begin{aligned}
\Psi_4 = & \frac{3 B^4 \lambda p^2 r^p}{8 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)} + \frac{B^4 \lambda \sqrt{p+1} p \sqrt{-p+1} r^p}{8 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)} + \frac{3 i B^3 \lambda p^2 r^2 p}{2 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)} \\
& + \frac{i B^3 \lambda \sqrt{p+1} p \sqrt{-p+1} r^2 p}{4 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)} - \frac{3 B^4 \lambda p r^p}{8 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)} - \frac{9 B^2 \lambda p^2 r^3 p}{4 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)} \\
& - \frac{3 i B^3 \lambda p r^2 p}{4 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)} - \frac{3 i B \lambda p^2 r^4 p}{2 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)} \\
& + \frac{i B \lambda \sqrt{p+1} p \sqrt{-p+1} r^4 p}{4 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)} + \frac{3 \lambda p^2 r^5 p}{8 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)} \\
& - \frac{\lambda \sqrt{p+1} p \sqrt{-p+1} r^5 p}{8 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)} - \frac{3 i B \lambda p r^4 p}{4 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)} + \frac{3 \lambda p r^5 p}{8 (B^6 r^2 + 3 B^4 r^2 r^2 p + 3 B^2 r^2 r^4 p + r^2 r^6 p)}
\end{aligned}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"